

Exercise Sheet 13

Discrete Mathematics by Hongfei Fu, 2022.11.10

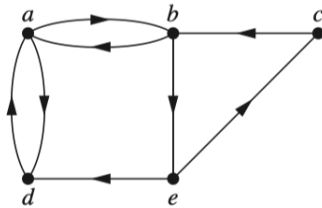
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1. ([R], Page 689, Exercise 2(a)(b)(d)) Does each of these lists of vertices form a path in the following graph? Which paths are simple? Which are circuits? What are the lengths of those that are paths?

a) a, b, e, c, b

b) a, d, a, d, a

d) a, b, e, c, b, d, a



Answer Area:

a) Yes. Simple path. The length is 4.

b) Yes. Circuit. The length is 4.

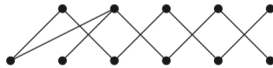
c) Not a path.

2. ([R], Page 689, 6) How many connected components does each of the graphs in Exercises 3-5 have? For each graph find each of its connected components.

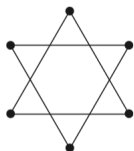
3.



4.



5.

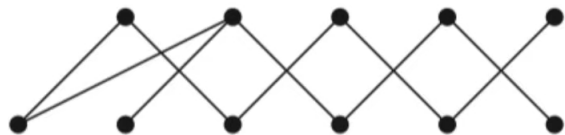


Answer Area:

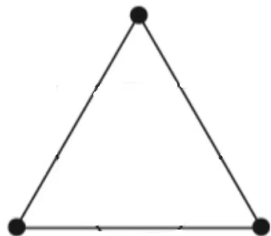
3. Three:



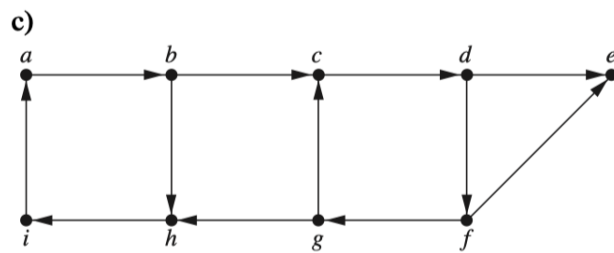
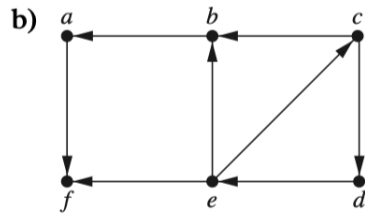
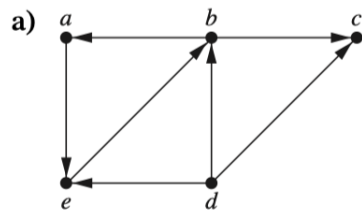
4. One:



5. Two:

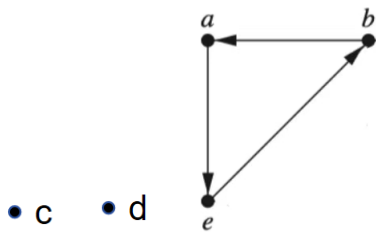


3. ([R], Page 690, 14(a)(b)(c)) Find the strongly connected components of each of these graphs.

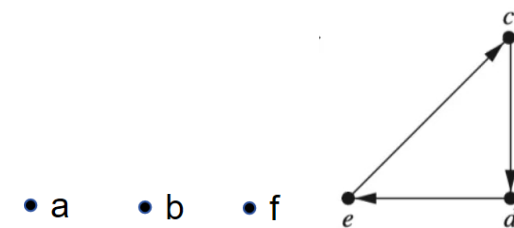


Answer Area:

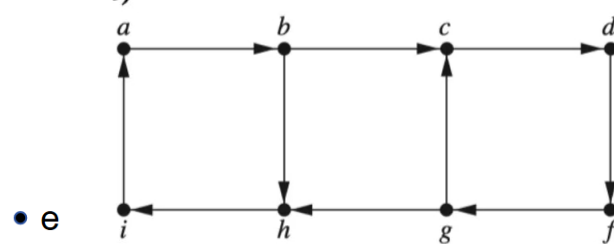
a)



b)



c)



4. ([R], Page 692, Exercise 63, **2pt**) Show that every connected simple graph with n vertices has at least $n - 1$ edges.

Answer Area:

- When $n = 1$, then every connected simple graph with one vertex, which is a vertex, has at least 0 edge.
- Suppose that every connected simple graph with n vertices has at least $n - 1$ edges. Then for every connected simple graph with $n + 1$ vertices, it can be seen as adding a new vertex v into the connected simple graphs with n vertices and $n - 1$ edges. Let u_1, u_2 be two vertices in the former graphs and are neighbours. Since the new graph is a simple graph, the degree of v is at least 2. And the two edges whose endpoint is v can replace at most one edge in the former graph. That is, remove edge $\{u_1, u_2\}$ and add $\{u_1, v\}$ and $\{v, u_2\}$.

So every connected simple graph with $n + 1$ vertices has at least n edges.

Therefore, according to the mathematical induction every connected simple graph with n vertices has at least $n - 1$ edges.

5. Show that a vertex c in the connected simple graph $\mathbf{G}$ is a cut vertex if and only if there are vertices u and v , both different from c , such that every path between u and v passes through c .

Answer Area:

- Suppose that vertex c in the connected simple graph $\mathbf{G}$ is a cut vertex.
Assuming that c is in a connected component $\mathbf{G}'$
 $\therefore c$ is a cut vertex and remove c only change $\mathbf{G}'$.
 $\therefore$ After removing c , $\mathbf{G}'$ contains two or more connected components. Assuming there are $\mathbf{G}_1, \mathbf{G}_2$.
 $\therefore \exists u$ in $\mathbf{G}_1$ and v in $\mathbf{G}_2$, such that u and v are not connected. But in original graph, they are connected.
 $\therefore$ There are vertices u and v , both different from c , such that every path between u and v passes through c .
- Suppose that there are vertices u and v , both different from c , such that every path between u and v passes through c .
Before removing c , u and v are connected and are in the same connected component. After removing c , u and v are not connected and are in two different connected components. Therefore the number of connected components increases.
So vertex c in the connected simple graph $\mathbf{G}$ is a cut vertex.